

Homework (Carolina Reaper)

- Q1.** A painter uses 5 different brush sizes. The sizes are 10, 15, 20, 25, and 30 mm. What is the *mean* brush size?
(A) 20 mm
(B) 21 mm
(C) 22 mm
(D) 23 mm
(E) 24 mm
- Q2.** A music band has 7 songs with the following lengths in minutes: 3, 4, 5, 6, 7, 8, 9. What is the *median* song length?
(A) 4 minutes
(B) 5 minutes
(C) 6 minutes
(D) 7 minutes
(E) 8 minutes
- Q3.** A sculptor creates 9 statues with the following heights in cm: 150, 160, 170, 170, 180, 190, 200, 210. What is the *mode* height?
(A) 150 cm
(B) 160 cm
(C) 170 cm
(D) 180 cm
(E) 190 cm
- Q4.** Two art galleries have the following number of paintings: Gallery A has 15, 20, 25, 30, 35 paintings and Gallery B has 10, 15, 20, 25, 30 paintings. Which gallery has a higher *mean* number of paintings?
(A) Gallery A
(B) Gallery B
(C) Both have the same mean
(D) Not enough information
(E) Cannot be determined
- Q5.** A musician practices for 4, 5, 6, 7, 8 hours a day. What is the *range* of practice hours?
(A) 1 hour
(B) 2 hours
(C) 3 hours
(D) 4 hours
(E) 5 hours
- Q6.** A painter has 3 canvases with the following areas in square meters: 10, 15, 20. What is the *median* area?
(A) 10 m²
(A) 12 m²
(A) 15 m²
(A) 18 m²
(A) 20 m²
- Q7.** A composer writes 5 songs with the following time signatures: 3/4, 4/4, 4/4, 5/4, 6/4. What is the *mode* time signature?
(A) 3/4
(B) 4/4
(C) 5/4
(D) 6/4
(E) No mode
- Q8.** Two music albums have the following number of tracks: Album A has 10, 12, 15, 18, 20 tracks and Album B has 8, 10, 12, 15, 18 tracks. Which album has a higher *median* number of tracks?
(A) Album A
(B) Album B
(C) Both have the same median
(D) Not enough information
(E) Cannot be determined

Open Ended Questions (FRQ)

- Q9.** A dance company has 6 performances with the following audience sizes: 200, 250, 300, 350, 400, 450. Calculate the *mean*, *median*, and *mode* audience sizes. Show all work and explain your answers.
- Q10.** A painter has 8 paintings with the following prices in dollars: 500, 600, 700, 800, 900, 1000, 1100, 1200. Calculate the *mean*, *median*, and *range* of the prices. Explain how these measures of central tendency help describe the data set.

Chef's Tips

- Ensure students use proper mathematical notation and show work for all calculations.
- Emphasize the importance of interpreting the results in context.
- Encourage students to compare and contrast the different measures of central tendency.